

- 4 The photograph shows a polychaete, *Sabellastarte spectabilis*, found in the Indian Ocean.



(Source: © LABETAA Andre/Shutterstock)

This animal can be kept in tanks in a laboratory.

This animal uses the fan to feed on organic particles suspended in sea water.

When the fan is touched, the fan is withdrawn into the tube by a reflex action.

The fan then extends to allow the animal to continue feeding.

A student observed that some of these animals withdrew their fans completely while other animals showed very little reaction to touch.

The student formed the following hypothesis:

the greater the number of touches, the less the animal reacts.

Plan an investigation to find evidence to support or reject this hypothesis.

- (a) Describe preliminary practical work that you might undertake to ensure your proposed method would provide quantitative results.

(3)

(b) Devise a detailed method, including how you would control and monitor important variables.

(8)



(c) Describe how your results should be recorded, presented and analysed in order to draw conclusions from your investigation.

(3)

[illegible]

(Total for Question 4 = 17 marks)

TOTAL FOR PAPER = 50 MARKS